

DSA (Abdul Bari) - Day 38

45-day plan | YouTube playlist course

Shortest Path - Dijkstra

Shortest paths from a source (non-negative weights).

Key concepts

- Greedy: always expand the closest unvisited node.
- Uses a priority queue; $O(E \log V)$.
- Fails with negative edges.

Key idea / example

Relax edges: if $\text{dist}[u] + w < \text{dist}[v]$, update $\text{dist}[v]$.

Today's tasks (1.5h)

- ☐ Implement Dijkstra
- ☐ Use a min-heap
- ☐ Explain the greedy step

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)